

ESM_1: Codebook and Composite Risk Index Construction

Online Resource accompanying the main manuscript "Quantifying Hidden Exploitation: Dual-Method Prevalence Estimates of Labour Exploitation among Domestic Workers"

1. Purpose and Contents

This Online Resource documents (a) the construction of the continuous composite risk index reported in Section 4.2 of the main manuscript, and (b) the codebook of key respondent-driven-sampling (RDS) variables used throughout the analysis. The composite risk index is a weighted sum across thirteen indicator categories drawn from the International Labour Organization's (ILO 2012, 2017) eleven indicators of forced labour, mapped to specific survey questions and response codings as set out in Table 1 below. The codebook of survey variables (Section 3 of this Resource) gives the variable names, source survey questions, and value codings for the RDS variables referenced in Section 3 of the main manuscript.

Detailed survey question wording is provided in ESM_2 (Survey Instrument). The three-step bootstrap procedure used to compute confidence intervals for the NSUM estimates is described in ESM_3. Sensitivity analyses, Bayesian model-assisted estimates, and the continuous- τ sweep are reported in ESM_4.

2. Composite Risk Index Construction

The composite risk index reported in Section 4.2 is a continuous variable computed as the sum of weighted contributions from eleven indicator categories drawn from the ILO conceptual framework (Categories 1 through 11 in Table 1 below). Each indicator category is anchored on the ILO classification of 'medium' indicators (whose presence individually is suggestive but not definitive of forced labour) and 'strong' indicators (any one of which alone is sufficient to indicate forced labour). Within-category weights are assigned according to the number of qualifying response items per category, such that the maximum contribution of any single category to the composite index is 0.09. The full index therefore ranges from 0 (no qualifying responses across any category) to 0.99 (maximum response on every qualifying question across all eleven categories).

Two further indicators (Categories 12 and 13 in Table 1) are reported as separate binary flags rather than included in the composite index. Category 12 (payment below the national living wage) is a binary indicator derived from the respondent's hourly pay (Q36); Category 13 (NRM-referral history) is a binary indicator of whether the respondent has been advised to enter the UK's National Referral Mechanism (Q80). Both indicators are reported separately in the main manuscript because each carries direct evidentiary status under UK policy independent of the composite-index construction.

Table 1. Composite Risk Index Construction. Mapping of ILO indicators of forced labour to survey questions, response codings, and per-category contribution weights. Categories 1–11 enter the continuous composite risk index (maximum value 0.99); Categories 12 and 13 are reported as separate binary indicators. Source: Authors' construction following ILO (2012, 2017).

Catego ry	ILO indicator (severity)	Source survey question(s)	Coding rule	Max weight
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1	Abuse of vulnerability (medium)	Q29 (5a1): Forced to do work against will Q51 (5d2-a): Required to perform additional unpaid tasks	Q29: 0 / 1 / 2 → 0.045 Q51: 0 → 0.045	0.09
2	Deception (medium)	Q45 (5c2): Forced, deceived or threatened into poor working conditions	Q45: 0 → 0.09	0.09
3	Restriction of movement (medium)	Q32 (5a4): Could not leave work premises Q46 (5c3): Stay permanently in household overnight	Q32: 0 / 1 / 2 → 0.045 Q46: 2 or 3 → 0.045	0.09
4	Isolation (medium)	Q65: Have friends outside work	Q65: 1 (No) → 0.09	0.09
5	Physical and sexual violence (strong)	Q44 (5c1): Experienced abuse, harassment, physical violence Q44 (5c1-b): Experienced sexual violence	Q44 (5c1): 0 → 0.045 Q44 (5c1-b): 0 → 0.045	0.09
6	Intimidation and threats (strong)	Q47 (5c4): Employer threatened or intimidated respondent	Q47: 0 / 1 / 2 → 0.09	0.09
7	Retention of identity documents (strong)	Q70 (5f2): Employer or agent withheld travel and identity documentation	Q70: 0 / 1 / 2 → 0.09	0.09
8	Withholding of wages (medium)	Q42 (5b7): Pay ever withheld by employer	Q42: 0 / 1 / 2 → 0.09	0.09
9	Debt bondage (medium)	Q39 (5b4): Earnings used to pay off debt to someone who helped find work	Q39: 0 → 0.09	0.09
10	Abusive working and living conditions (medium)	Q48 (5c5): Employer verbally abused Q63 (5d10): Lack of privacy at work Q72 (5f4): Safe and healthy working environment Q76 (5f8): Access to good-quality food, water, sanitation	Q48: 0 / 1 / 2 → 0.0225 Q63: 0 / 1 / 2 → 0.0225 Q72: 2 or 3 → 0.0225 Q76: 1 / 2 / 3 → 0.0225	0.09

11	Excessive overtime (medium)	Q16 (3b): Weekly hours worked Q61 (5d8): Weekly rest longer than 24 hours consecutively Q62 (5d9): Worked overtime felt to be excessive	Q16: 6 (45–70 hrs) or 7 (70+ hrs) → 0.03 Q61: 2 or 3 → 0.03 Q62: 0 / 1 / 2 → 0.03	0.09
12	Payment below national living wage (separate binary)	Q36 (5b1): Hourly pay before tax	Q36: 0 / 1 / 2 (below £10.42/hr) → 1	Report ed separa tely
13	NRM-referral indicator (separate binary)	Q80 (5f12): Respondent or someone known has been advised to enter the National Referral Mechanism	Q80: 0 or 1 → 1	Report ed separa tely

3. Binary ILO-Aligned Indicators (Section 4.3)

Section 4.3 of the main manuscript reports prevalence estimates for five binary ILO-aligned indicators rather than the full thirteen-category composite. These five binary indicators are constructed directly from specific survey questions, in some cases overlapping with the categories that feed the composite index but in some cases drawing on questions outside the thirteen-category system. Table 2 below documents the mapping. The constructions follow the comparable-indicator specifications agreed with Caroline Emberson in November 2024 to ensure direct comparability between the egocentric RDS estimates and the alter-centric NSUM estimates fielded on the same respondents.

Table 2. Five Binary Indicators Reported in Section 4.3. Each binary indicator equals 1 if any of the listed source questions returns an affirmative response under the coding rule. Source: Authors' construction.

Binary indicator	ILO classification	Source survey question(s)	Coding rule
Document withholding	Strong	Q70 (5f2): Identity documentation withheld	Q70: 0 / 1 / 2 → 1
Threats and abuse	Strong (Cat 5+6)	Q45 (5c2): Forced into poor conditions Q47 (5c4): Threatened or intimidated Q48 (5c5): Verbally abused	Any of: Q45=0; Q47=0/1/2; Q48=0/1/2 → 1
Pay issues	Medium (Cat 8+9)	Q39 (5b4): Debt bondage Q42 (5b7): Pay withheld	Either Q39=0 or Q42=0/1/2 → 1
Excessive hours	Medium (Cat 11)	Q61 (5d8): Weekly rest	Either Q61=2/3 or

		Q62 (5d9): Excessive overtime	Q62=0/1/2 → 1
Limited access to help	Auxiliary	Q78 (5f10): Access to support (reverse-coded)	Q78 indicating no access → 1

4. Codebook of RDS Variables

Table 3 below lists the key respondent-driven-sampling variables referenced in Section 3 of the main manuscript. These variables are used to construct the RDS sampling weights (RDS-I, RDS-II/Volz-Heckathorn, and RDS-SS/Gile), to characterise the recruitment-chain structure for diagnostic and bootstrap purposes, and to estimate the personal network sizes that feed the NSUM construction. Additional demographic and outcome variables — including nationality cluster, recruitment wave, and the five binary indicators of Section 3 above — are documented in the main manuscript's Section 3.3 (Sample Characteristics).

Table 3. Key RDS Variables. Variable definitions and source survey questions for the variables used to construct RDS sampling weights and network-size estimates. Source: Authors' construction.

Variable	Type	Definition	Source
id	Numeric	Unique numeric identifier for each respondent in the RDS sample.	—
recruiter.id	Numeric	id of the respondent who recruited the focal respondent. Value of -1 indicates a seed (initial sample member with no recruiter).	—
referral_1 through referral_5	Numeric	ids of up to five respondents recruited by the focal respondent. Missing values indicate fewer than five successful referrals.	—
q13 (2f)	Numeric (count)	Personal network size: number of other domestic workers the respondent personally knows and has contact details for. Used as the degree variable in RDS weighting and as the NSUM denominator.	Q13 (Section 2f of the instrument)
1a_eligibility	Binary	Whether the respondent has worked as a domestic worker in the past 12 months. Eligibility filter; all respondents in the analytical sample have 1a_eligibility = 1.	Q1a
wave	Numeric (0, 1, 2, ...)	Recruitment wave. Wave 0 indicates a seed; wave k indicates the respondent was recruited by a wave-(k-1) respondent.	Derived from recruiter.i

			d
nationality cluster	Categorical (4 levels)	Nationality cluster used for stratified analyses. Levels: Filipino, Latinx, British, Other.	Q3, Q5

5. Data Availability and Reproducibility

The raw survey data (with personally identifying information removed) and all analysis code used to construct the composite risk index and the binary indicators above are available in the GitHub repository at <https://github.com/smoser11/DWeMSinUK>. The composite-index construction is implemented in R/data_processing/02-data_preparation.R. The five binary indicators are constructed in the same script by the `create_comparable_indicators()` function, which implements the specifications agreed with Caroline Emberson in November 2024 (see Notes/2024-11-05 in the repository). An anonymised view-only mirror of the analysis code is available on the Open Science Framework at [OSF URL TO BE INSERTED].

Researchers wishing to replicate the construction of the composite risk index can recover the per-category contribution columns directly from the processed data file `data/processed/data_full.csv`, which contains `category_1` through `category_13` as named columns alongside the `composite_risk` total. The pairwise mapping of category columns to ILO indicators follows Table 1 above; the mapping of binary-indicator names to source columns follows Table 2 above.